

A fixed-representative partial reduction for Question 1 of arXiv:1912.03994

Run `fa_banach.001`; source arXiv:1912.03994

Status

partial result.

Cuth–Dolezal–Doucha–Kurka ask whether there is a continuous map $\Phi : \mathcal{P}_\infty \rightarrow \mathcal{B}$ such that X_μ is linearly isometric to $X_{\Phi(\mu)}$ for every $\mu \in \mathcal{P}_\infty$. This packet gives a positive answer on every stratum where an independent dense representative sequence is fixed in advance.

Source Questions

The relevant source questions appear in arXiv:1912.03994, pages 20–21.

for every $\sum_{k=1}^\infty \alpha_k e_k \in c_{00}$ and every $\mu \in \mathcal{D}$. □

To summarize, we obtained an optimal reduction from $\mathcal{B}$ to $SB_\infty(X)$ and from $SB_\infty(X)$ to $\mathcal{P}_\infty$. However, our reduction from $\mathcal{P}_\infty$ to $\mathcal{B}$ seems not to be optimal, so one is tempted to ask the following.

Question 1. Does there exist a continuous mapping $\Phi : \mathcal{P}_\infty \rightarrow \mathcal{B}$ such that for every $\mu \in \mathcal{P}_\infty$ we have $X_\mu \equiv X_{\Phi(\mu)}$?

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Note that a positive answer to Question 1 would imply a positive answer to Question 2 and that a sufficient condition for a positive solution of Question 2 is provided by Proposition 2.15.

Question 2. Let X be an isometrically universal separable Banach space and let τ be an admissible topology on $SB(X)$. Does there exist a Σ_2^0 -measurable mapping $\Phi : \mathcal{P}_\infty \rightarrow (SB(X), \tau)$ such that for every $\mu \in \mathcal{P}_\infty$ we have $X_\mu \equiv \Phi(\mu)$?

Question 1 asks for a continuous reduction $\mathcal{P}_\infty \rightarrow \mathcal{B}$. Question 2 asks for the corresponding Σ_2^0 -measurable reduction from $\mathcal{P}_\infty$ to $SB(X)$ with an admissible topology.

Definitions

Let V be the rational vector space with basis $(e_n)_{n \in \mathbb{N}}$, as in the source paper. For $\mu \in \mathcal{P}$, write X_μ for the completion of (c_{00}, μ) modulo its null space, and write $[v]_\mu$ for the image of $v \in V$ in X_μ .

Fix a strictly increasing sequence $s = (s_j)_{j \geq 1}$ in $\mathbb{N}$. Let $\mathcal{P}_s \subseteq \mathcal{P}_\infty$ be the class of all μ such that:

1. for every nonzero finite rational scalar family (a_j) ,

$$\mu\left(\sum_j a_j e_{s_j}\right) > 0,$$

so the vectors $[e_{s_j}]_\mu$ are linearly independent;

2. the closed linear span of $\{[e_{s_j}]_\mu : j \geq 1\}$ is X_μ .

Proof Intuition

The source paper already gives a Σ_2^0 -measurable reduction by choosing, for each μ , the first coordinates that remain linearly independent in X_μ . The discontinuity can only enter through that coordinate-selection step. If the representative coordinates are fixed beforehand, no selection remains: the output norm is just the restriction of μ to that fixed subsequence. Continuity is then immediate from the coordinate topology on $\mathcal{P}$.

Theorem

Theorem 1. *For every strictly increasing sequence $s = (s_j)$ in $\mathbb{N}$, the formula*

$$\Phi_s(\mu) \left(\sum_{j=1}^m a_j e_j \right) = \mu \left(\sum_{j=1}^m a_j e_{s_j} \right), \quad a_j \in \mathbb{Q},$$

defines a continuous map $\Phi_s : \mathcal{P}_s \rightarrow \mathcal{B}$, where $\mathcal{P}_s$ has the relative topology inherited from $\mathcal{P}_\infty$. Moreover

$$X_{\Phi_s(\mu)} \equiv X_\mu$$

linearly isometrically for every $\mu \in \mathcal{P}_s$.

Proof. Fix s and $\mu \in \mathcal{P}_s$. The formula defines a pseudonorm on V because it is the pullback of the pseudonorm μ under the fixed rational linear injection $T_s(e_j) = e_{s_j}$. The first defining property of $\mathcal{P}_s$ says exactly that no nonzero vector of V is sent to a μ -null vector; hence $\Phi_s(\mu)$ is a norm and belongs to $\mathcal{B}$.

For continuity, let $v = \sum_{j=1}^m a_j e_j \in V$. Then

$$\Phi_s(\mu)(v) = \mu(T_s v).$$

Since $T_s v \in V$ is fixed, the map $\mu \mapsto \mu(T_s v)$ is one of the coordinate maps defining the topology of $\mathcal{P}$. Therefore every coordinate of Φ_s is continuous, which is precisely continuity as a map into $\mathcal{B} \subseteq \mathcal{P}$.

Finally define $J_\mu : V \rightarrow X_\mu$ by $J_\mu(e_j) = [e_{s_j}]_\mu$. For every $v \in V$,

$$\|J_\mu v\|_{X_\mu} = \Phi_s(\mu)(v).$$

Thus J_μ descends to an isometric embedding of $X_{\Phi_s(\mu)}$ into X_μ . Its range contains the span of $\{[e_{s_j}]_\mu : j \geq 1\}$, which is dense in X_μ by the second defining property of $\mathcal{P}_s$. Hence the extension of J_μ is onto and is a linear isometry. $\square$

Corollary 2. *Let $A \subseteq \mathbb{N}$ have infinite complement $S = \{s_1 < s_2 < \dots\}$. On the stratum of all $\mu \in \mathcal{P}_\infty$ for which the coordinates $\{e_i : i \in A\}$ vanish and the quotient vectors $\{[e_{s_j}]_\mu : j \geq 1\}$ are linearly independent, Question 1 has a positive answer via Φ_s .*

Proof. Under the stated hypotheses, every coordinate outside S is zero in the quotient and every coordinate inside S belongs to the fixed representative sequence. Hence the span of $[e_{s_j}]_\mu$ is dense in X_μ , and the theorem applies. $\square$

Relation to Question 2

The source paper proves a Σ_2^0 -measurable reduction from $\mathcal{B}$ to $SB_\infty(X)$ for any admissible topology on $SB(X)$. Composing that reduction with the continuous map Φ_s gives the corresponding relative positive answer to Question 2 on each stratum $\mathcal{P}_s$.

Verification and Limitations

The proof is purely formal and uses only the definitions of the source coding spaces. It does not solve the global problem because it assumes that the representative sequence s is fixed on the whole domain. Thus the remaining obstruction is precisely the source paper's global selection problem: choosing such an independent dense sequence continuously as μ varies over all of $\mathcal{P}_\infty$.

Local index searches for arXiv:1912.03994, Question 1, $\mathcal{P}_\infty$, $\mathcal{B}$, continuous reduction, and admissible topologies found no prior packet for this fixed-representative partial result. Web searches on 2026-06-16 found the source paper and the sequel arXiv:2204.06834, but no exact global answer to Question 1 or Question 2.

References

References

- [1] M. Cuth, M. Dolezal, M. Doucha, and O. Kurka, *Polish spaces of Banach spaces*, arXiv:1912.03994.
- [2] M. Cuth, M. Dolezal, M. Doucha, and O. Kurka, *Polish spaces of Banach spaces. Complexity of isometry and isomorphism classes*, arXiv:2204.06834.